



Signsprout

Learn your first ASL signs with *your own two hands*.

Sprout shows each sign face to face, guide hands sit where yours are, and every sign grows your garden. Five minutes a day.

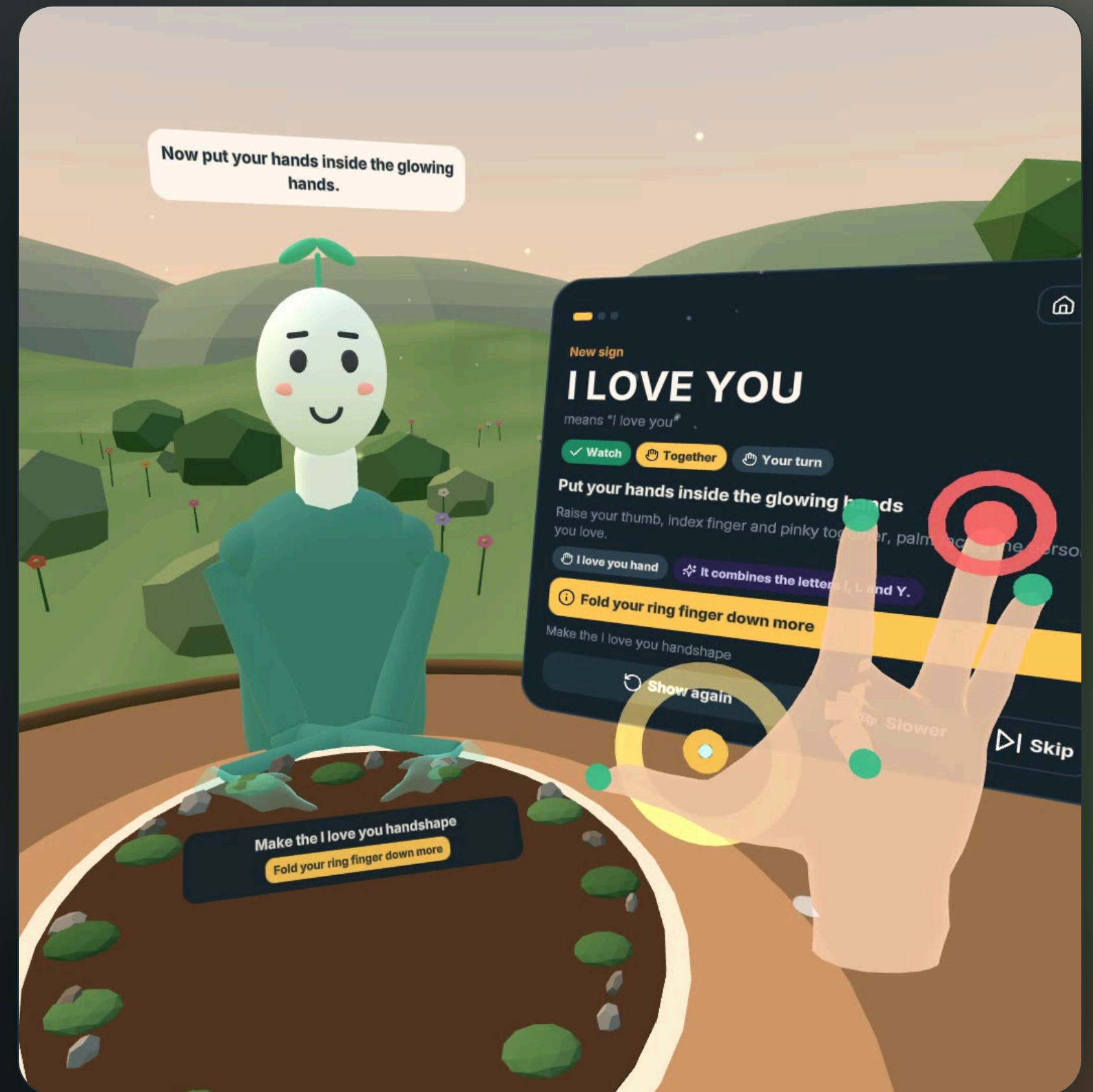
Productivity

New Experience

Hands only · Seated

Meta VR Start Developer Competition 2026

Made by **Shivam Gupta**



— THE PROBLEM

Nine out of ten deaf babies are born to *hearing parents*.

Most have never signed. The first months are when their baby most needs an accessible language, and when parents have the least time.

90%+

of deaf children are born to hearing parents

1 in 4

families with deaf children regularly sign at home (22.9%)

6 months

exposure before this age leads to age-level ASL vocabulary

— THE INSIGHT

Videos make you do the hardest part *in your head*.

1. You have to mirror everything

The signer faces you, so their right hand is on your left. Hearing non-signers copying a face-to-face model made errors on **24.3%** of sideways movements. In a second study, a model that matched their mirroring cut those errors from **20.6% to 0.9%**.

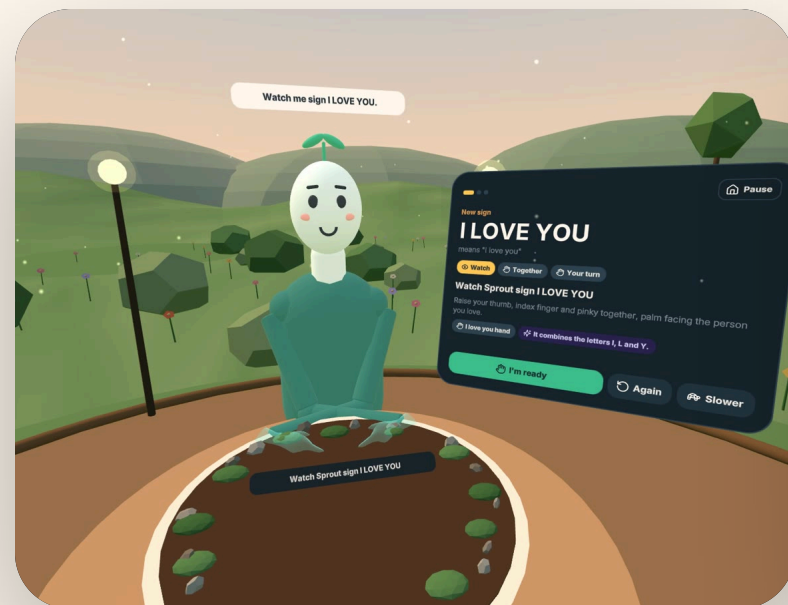
2. Nobody checks your hands

A video can't see that your ring finger is up or your hand is at your cheek instead of your chin. Parents practise alone, late at night, and never know if they got it right.

A headset can help with both: put guide hands *exactly where yours are*, and read 25 joints per hand.

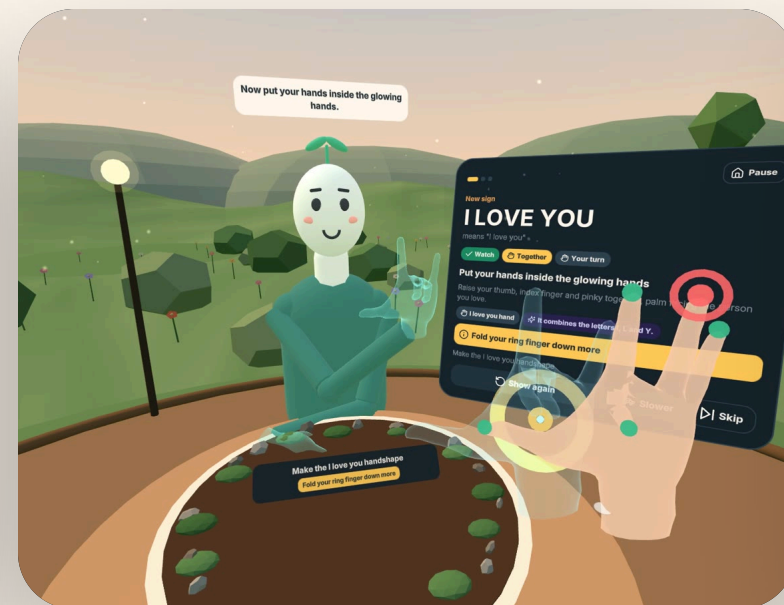
— HOW IT WORKS

Four steps, about a minute a sign.



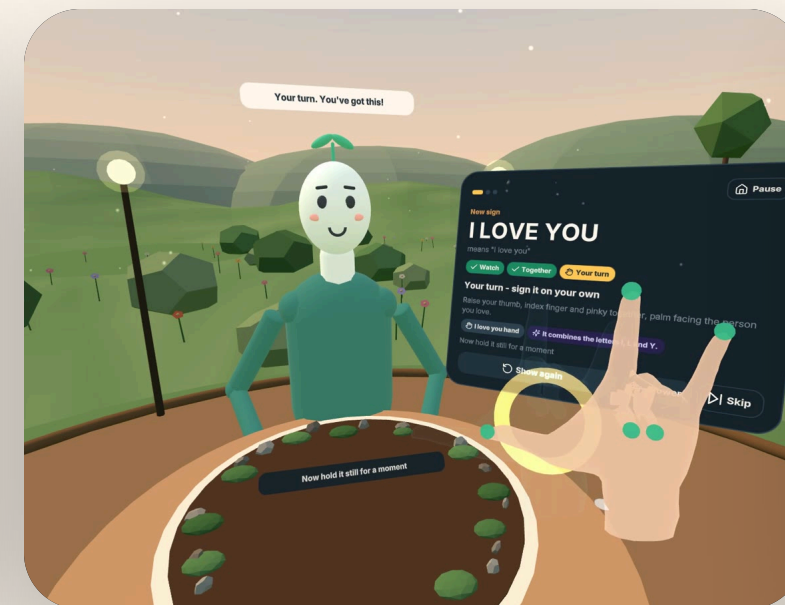
1 Watch

Sprout shows the sign face to face, mirrored so it's easy to copy.



2 Together

Glowing guide hands appear on your hands. Step inside and sign along.



3 Your turn

From memory. It checks handshape, then place, then movement, then the final handshape, and says what to fix.



4 Celebrate

A new plant sprouts in your garden.



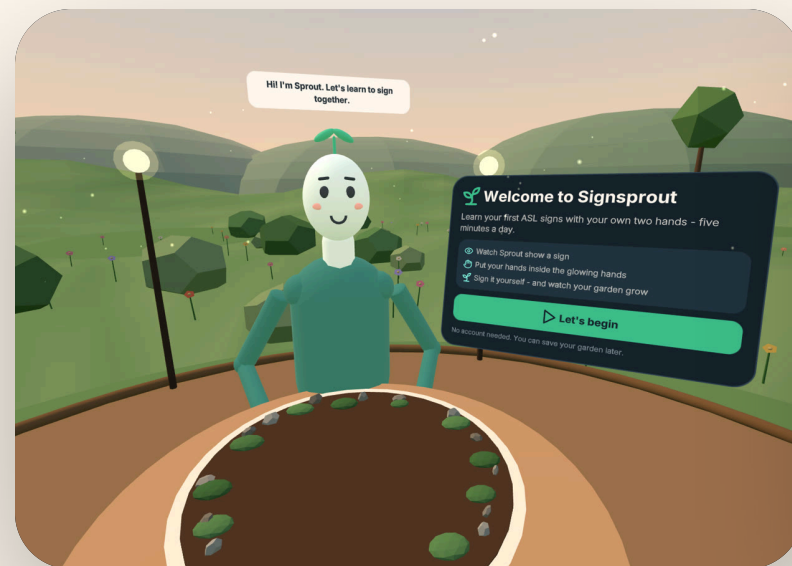
— FEEDBACK IN PLAIN WORDS

"Fold your ring finger down more."

25 joints per hand, read every frame. Each finger gets a status you can see (colour *and* shape) and a hint in plain words.

— THE FIRST FIVE MINUTES

No wall of text. No account. *Three real signs.*



Poke "Let's begin"
with a fingertip



Hold up your hands
Sprout sees them



Pick your signing hand
every sign mirrors to match



Touch your chin
calibrates face landmarks

Then HELLO, THANK-YOU and I-LOVE-YOU, and the first three plants in your garden.

— A REASON TO COME BACK

Your garden *is* your progress.

- Plants grow as signs move into long-term memory (spaced repetition graded by how the attempt went, not by self-rating).
- They droop when a review is due, and a poke replays the sign.
- Five-minute sessions, streaks, a daily goal.
- Guide hands fade after two unaided successes, so skill transfers.

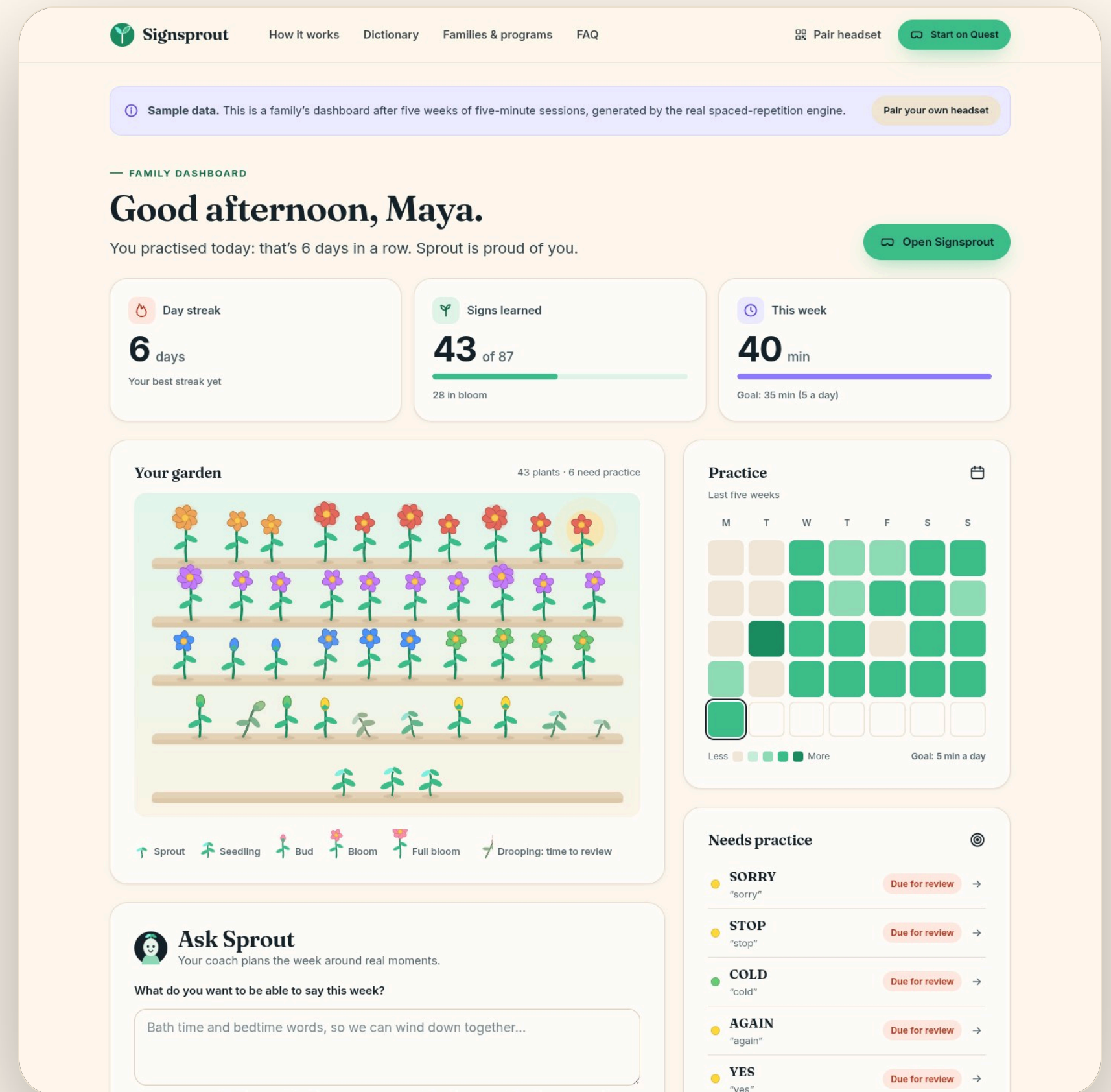


— AGENTIC, BUT GROUNDED

Sprout plans your week around *real moments*.

"Bath time and bedtime words." The coach uses Gemini function calling over the sign catalogue and the learner's progress. It can only choose signs Signsprout can teach and check, and the plan appears in the headset.

- Phone pairing with a 6-character code: no typing in VR
- Family dashboard, 3D dictionary, read-only share links
- Runs on Supabase; the public preview works fully offline



Signs are data, *not video*.

```
// THANK-YOU, written the way linguists describe signs
oneHanded(
  { id: 'thank-you', gloss: 'THANK-YOU' },
  { shape: 'open-B', at: 'chin',
    contact: 'fingertips', palm: 'in' },
  { to: { at: 'chin', offset: [0.03, -0.12, -0.22],
    palm: 'up-in' }, path: 'arc' },
);
```

One definition drives:

- Sprout's animation and the first-person guide hands
- The dotted guide path and the written instructions
- The step-by-step checker
- Left-handed mirroring, automatically

50 handshapes solved with forward and inverse kinematics from a real Quest hand capture. No training data needed per sign.

— RELIABILITY

Checked one part at a time.

- 1 · Shape
- 2 · Place (with a brief hold)
- 3 · Movement
- 4 · Final shape

TESTED WITH A SIMULATED LEARNER (SYNTHETIC HANDS, NOT YET REAL-USER DATA)	RESULT
All 87 signs performed correctly, right- and left-handed	100% accepted
Wrong handshape, wrong place or missing movement	0 false accepts
8° joint jitter and 4 mm position noise	98.5% accepted
Signing at 0.6× and 1.5× speed	100% accepted
One finger in the wrong position (a subtle slip)	94% caught

packages/signkit/test and scripts/sim-*.ts. The simulated learner is built from the same hand model as the checker, so these results show consistency, not accuracy on real hands. On-device testing with real signers is next.

— BUILT FOR THE HEADSET

Hands only. Seated. *A link, not an install.*

Hands-first

Every button (about 2.8–4.5 cm tall) works with a fingertip poke, a pinch ray or look-to-select. Far rays switch off while you sign, so a pinch-like handshape doesn't click by accident.

Airplane-seat test

The stage recenters on your head, the table follows your eye height, everything sits within two feet. Pause and resume follow the headset.

Fast and light

Meta Immersive Web SDK 1.0 and WebXR Hand Input. Procedural, instanced scene; the garden draws ~90 plants in a handful of draw calls.

— ACCESSIBLE BY DEFAULT

Designed for every body that wants to learn.

- Captions for every prompt, optional voice
- Left-handed mode: every sign mirrors, the panel moves left
- "Limited finger range" and "Continue with one hand"
- Adjustable strictness and guide speed
- High contrast, calm motion, colour plus shape

Respect for the Deaf community

- A practice partner, never a translator
- The site links to Deaf-led resources and Deaf Mentor programs
- We compared each sign with Handspeak and Lifeprint. No Deaf signer has reviewed Signsprout yet; paid review by Deaf signers comes before launch
- Hand data never leaves the headset; facial grammar is mentioned as a tip, not scored

— MARKET

Start where the need is sharpest.
Grow with every learner.

Beachhead

5,600

new US hearing families of deaf babies every year; ~28,000 with a child under five, several learners each

Expansion

#3

ASL is the third most-studied language in US colleges (107,899 enrolments); 2.8% of US adults sign

Proof of demand

2.5M

learners on Lingvano with 2D video alone. Duolingo still has no ASL course.

— BUSINESS MODEL

Planned: families of deaf babies *never pay.*

Free

\$0

First words, Mealtime and Family units, fingerspelling and numbers, the garden

Family

\$59 /year

or \$7.99/month. All signs, AI coach, sync, dashboard, up to 4 family members

Family Pass

\$0

Planned: free for families of deaf children under 3, funded through partner programs

Programs

\$39 /family

per year: seats, consented progress reports, headset lending kits

Costs stay tiny: static hosting, a managed database, on-device hand tracking.
Under \$0.05 per active family per month at 10,000 families (estimate). Non-dilutive funding fits: NSF and ED SBIR.

— WHY SIGNSPROUT WINS

The only one that puts guide hands *on yours.*

	SIGNSPROUT	NVIDIA SIGNS	LINGVANO	POPSIGN	QUEST ASL APPS
Checks your own hands	3D, 25 joints/hand	2D webcam	Self-view only	Phone camera	Mostly letters
Guide hands on yours	Yes	No	No	No	No
Says exactly what to fix	Per finger and parameter	Real-time, 2D	No	No	Limited
Family curriculum + spaced repetition	Yes	No	Yes	Game	No

Moat: an engine where a new sign is a few lines of data, trust we plan to earn through paid Deaf co-leadership with a veto over content, and distribution through early-intervention programs that consumer apps rarely reach.

— WHAT'S NEXT

From competition build to every family that needs it.

Nov 2026

Competition build

87 signs, coach, pairing, companion site

Dec 2026

Open web beta

Testing with real signers, paid Deaf reviewers, three pilot programs

Q1 2027

Meta VR Store

Family plan; every sign reviewed by Deaf signers

Q2 2027

Deeper practice

Facial-grammar lessons, two-person practice

H2 2027

More languages

Phone-camera mode; BSL and ISL with local Deaf partners



Five minutes tonight.
A first sign tomorrow.



Open in the Quest browser

shi1720.github.io/meta-vr/app

Made by Shivam Gupta